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like the person who wrote in Ostia *pugízō pántas toùs epitoikhográphous* 'I will screw all writers on walls' (Rea 1979).

Proper names constitute the only universal category of graffiti. They are often followed by "I was here" (*paregenómēn*), "wrote" (*egráphē*), or similar phrases. The writing of a proper name could have a religious connotation, as one can see in the many *mnḗsthē* ('remember') graffiti from Dura-Europos (Baird 2011:56) or Aphrodisias (Chaniotis 2011:193), for instance, as well as in the signatures in El Kanais in Egypt (Mairs 2011:159).

Graffiti of sexual insult and obscenity are found on the rocks of Hymettos (Langdon 2004), as well as on the walls of Pompeii (Kaimio 1979:170), and Aphrodisias (Chaniotis 2011:204), for instance. A very usual word written on walls is *katápugon* or *katapúgōn*, a derogatory term for passive homosexuals (→ Aischrology).

Another broadly distributed type is that of "kalós graffiti", which celebrate the beauty of a young man and thus are considered to be graffiti of paederastic acclamation. There are more than 700 "kalós graffiti" in Attica alone, mostly on vases, but also on walls (Langdon 2004:202). Sometimes the adjective *kalós* 'beautiful' is followed by other adjectives, like *hēdús* ('sweet'), *eúkharis* ('charming'), etc. (Garlan-Masson 1982:17).

Graffiti on walls or rocks could have been made by the inhabitants of a city, a town, or a village (i.e., Athens, Ephesos, etc.), as well as by travellers (i.e., in Hymettos, in Attica). They should be read in context, and intertextual interaction with other graffiti should be considered. For instance, many signatures are attached to texts already written on the rock, showing a more complex pragmatic meaning.

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Beatriz Moncó-Taracena

Grammaticalization

→ Syntactic Change

Grassmann's Law

"Grassmann's Law" (Grassmann 1863, cf. Collinge 1985:47–61 for a detailed survey of initial reaction and subsequent debate) is a phonological rule within the history of Greek that describes the dissimilatory (→ dissimilation) loss of → aspiration in one of two non-adjacent aspirated consonants of a word. Typically, the first consonant (and the one most often affected) occurs at the beginning of the word, and the second at the onset of the following (open) syllable: e.g. Proto-Gk. **hékhō* > Gk. *ékhō* 'I have, hold', Proto-Gk. **phakhús* > Gk. *pakhús* 'thick'. When the two aspirates are tautosyllabic, the second loses its aspiration: e.g. Proto-Gk. **hékhsō* > Gk. *héksō* 'I will have'. (Additional examples and discussion in Lejeune 1972:56–58, Rix 1992:97, Sihler 1995:142–144.) Like the identically-named process within the history of Sanskrit, Grassmann's Law was discovered by the mathematician and linguist Hermann Grassmann (1809–1877).

Alternations arising from Grassmann's Law sometimes remain intact, especially with /h/ and /th/: hence (for /h/) Gk. *ékhō* beside *héksō* (above). In most other cases, → analogy has removed such alternations: thus the comparative of *pakhús* (above) is *pássōn* 'thicker', maintaining the onset /p-/, vs. the expected result ***phássōn* (with Grassmann's Law). (Compare *takhús* 'fast' vs. comparative *thắssōn*, with the Grassmann's Law alternation maintained.) Similarly, aspiration is analogically maintained in verbs formed with the aorist passive suffix *-thē-* (→ Aorist Formation), as in *ekhúthē* 'it was poured', vs. unattested ***ekúthē* (if Grassmann's Law had applied).

Grassmann's Law applied to the Proto-Gk. outcomes of PIE **s-* (> Proto-Gk. **h-*) and the PIE voiced aspirates **bh, *dh*, etc. (> Proto-Gk. **ph, *th*, etc.) (→ Indo-European Linguistic

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Background). Grassmann's Law in Greek is thus independent of Grassmann's Law in Sanskrit, which applied to the PIE voiced aspirates themselves. Indeed, dissimilation of aspiration has good typological parallels: cf. "Katupha's Law" in Makhuwa (a Bantu language), which dissimilates the first of two voiceless aspirated stops in successive syllables (Dimmendaal 2011:31). On Grassmann's Law (like other processes of "dissimilation at a distance") as perceptual hypercorrection, see Ohala (2003:678, 680).

The precise dating of the process remains somewhat controversial. But there are now good arguments based on Mycenaean Greek suggesting that the process was post-Mycenaean: e.g., the → hiatus in the compound *ko-to-no-o-ko* 'land-holding' implies /ktoino-hokhos/, in which the second member (based on Proto-Gk. **hekh-* 'have, hold', cf. above) retains its initial aspiration, without having undergone Grassmann's Law. (See Lejeune 1972:57, Plath 1987, Garrett 2006:141, Thompson 2010:191.) Dialectal double-aspirate forms of the type Arc., early Att. *pharthenos* (vs. literary Att. *parthénos* 'maid, virgin') – in this case, a word that very likely lacked an initial aspirate originally (Beekes 2010:1153, s.v.) – probably reflect late assimilations rather than pre-Grassmann's Law survivals (Buck 1955:60).

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Brent Vine

Greek and Anatolian Languages

1. Introduction

The Anatolian languages (< Greek *anatolḗ* 'the east') include Palaic and Cuneiform Luwian (2nd millennium BCE), Hieroglyphic Luwian (2nd–1st millennium BCE), Lycian and Lydian (1st millennium BCE), the poorly attested Luwian languages of Southern Anatolia – Carian, Pisidian and Sidetic, and Hittite (heavily attested from ca 1570–1200 BCE). Until the unearthing of the imperial archives at Hattuša (modern Boğazköy), knowledge of this family was limited to oblique references and almost mythical tales, such as the hapless biblical Uriah, the epics of Troy, or the occasional settlement, place name, custom, language, or divinity perceived as 'non-Greek'. An extensive body of scholarship has arisen in the last 100 years to address relations between Greece and Anatolia, with the result that several of these issues have been resolved: *Miletus* has been identified with *Millawanda, Ephesus* with *Apašaš, Wiluša* with (*W*)*ilios*, (used twice as often as 'Troy' in the *Iliad*). While the lively debate over the location of *Aḫḫiyawa* appears to have been settled (Melchert forthcoming; Güterbock 1984), other disputes between Anatolian and Aegean specialists continue unabated and hotly contested.

The evidence indicates that Anatolia both geographically and culturally formed a bridge between the West (Greece) and the cultures of the Middle East, with Anatolia influencing virtually every facet of Greek life – from the diffusion of peoples – the Trojan War affecting the "entire ethnography of the Mediterranean" (Malkin 1998:3), to warfare, pottery, textile and weaving technology, the area of myth, religion and ritual, and even athletics – the Olympics being "neither new nor specifically Greek . . . [but] . . . fully present in the Hellado-Anatolian orbit of the second millennium before the common era" (Puhvel 2002:70).

For most of their interaction, the influence was east to west (but see Finkelberg 2005:70). The tide turned toward the end of the Bronze

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the linkage of primary to secondary sources, and, finally, the registration of multiple interpretations attested on commentaries and reviews to texts with possible variants.

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Federico Boschetti

Cowgill's Law

'Cowgill's Law' refers to one or more phonological processes (→ Phonological Change) describing the development of PIE **o* to Gk. /u/ in restricted environments. (In all other contexts, PIE **o* > Gk. /o/). An influential statement is that of Sihler (1995:42): "**o* from any source > *u* in Greek between a labial and a resonant"; similar formulations were first proposed by the American Indo-Europeanist Warren Cowgill (1929–1985) (Cowgill 1965[2]:156–157). This rule was designed to account for the unexpected *u*-vocalism in Greek words with a source in PIE **o*, which had often been noted as problematic in earlier treatments such as that of Lejeune (1972:190 n.3). Examples: Gk. *núx* 'night' < PIE **nok^{w}t-s*, cf. Lat. *nox* (the 'labial' in this case being the PIE → labiovelar consonant **k^{w}*, still intact at the time of Mycenaean Greek (→ Mycenaean Script and Language)); and 1 pl. pres. Gk. *stórnumen* 'we spread' (and other 1 pl. forms based on similar nasal stems), with sequence *-nu-m-* < **-no-m-* < PIE **-nh$_3$-m-* (where **-nh$_3$-* is a nasal suffix and *-m-* is the beginning of the personal ending).

It has been argued, however, that Sihler's statement is too broad: numerous counterexamples that cannot be explained analogically, such as Att. *mónos* (Ion. *moûnos*) 'alone' < Proto-Gk. **monwo-*, and many other isolated forms in **wor-, *wol-, *mor-, *mol-, *por-, *pol-, *pon-*, etc., do not undergo such a change (Vine 1999:563–564). According to this argument, a 'Cowgill's Law proper' should be restricted to the following environments: (1) */noK^{w}/ (where K^{w} = any PIE labiovelar) and possibly its mirror image */K^{w}on/ (cf. *núx* above for the first; examples for the second are less clear-cut, but may include *gunḗ* 'woman' if from **g^{w}on-*, cf. Luw. *wāna-* 'woman'), as well as (possibly) */noK$^{(h)}$w/ (where K = any velar); (2) */nom/ (cf. *stórnumen* above), but not its mirror image */mon/. Additional forms with unexpected /u/ < PIE **o* (otherwise accounted for by the overly-general version of Cowgill's Law) should instead, in this view, be explained by other processes, namely (3) */oly/ > Gk. /ull/, e.g. PIE **skolH-ye/o-* > **skol-ye/o-* (with regular laryngeal loss (→ Laryngeal Changes)) > Gk. *skúllō* 'flay, rend', cf. Hitt. *iskalla-* 'tear' (Jasanoff 2003:78), and (4) */Cwo-/ > Gk. /Cwu-/ (where C = any consonant), e.g. Gk. *sū́rō* 'drag, sweep away' < **twor-ye/o-*, beside zero-grade **twr̥-ye/o-* in Gk. *saírō* 'sweep'. As a result, the term 'Cowgill's Law' is sometimes used loosely with reference to the entire complex of processes governing the development of (PIE or Proto-Gk.) **o* to (pandialectal Gk.) /u/.

These processes are distinct from dialectally-restricted developments of PIE **o* that have also led to sporadic *u*-vocalism, often where **o* was adjacent to a labial consonant: e.g. Lesb., Arc. *stúma* 'mouth' (vs. Att.-Ion. *stóma*), cf. the Mycenaean Gk. boonyms *tu-ma-ko* /*Stumargos*/

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and *to-ma-ko* /*Stomargos*/ 'White-mouth' (both attested at Knossos). For some material of this sort, such as the preposition and verbal prefix Lesb., Thess., Arc., Cypr. *apu*, Myc. *a-pu* vs. Att.-Ion. *apó* 'from', as well as for some items without variation sometimes thought to result from a generalized form of Cowgill's Law, e.g. *prumnós* 'endmost' if from **promno-* (Forssman 1965), it may be preferable to assume archaic → ablaut variants in **u*, which have been reconstructed for some → particles, e.g. **pru-* (beside **pro*), cf. Gk. *diaprúsios* 'penetrating' beside *diapró* 'thoroughly' (Dunkel 1992:161n.32).

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Crasis

Crasis (*krâsis*, a 'mixture' of two → vowels) as a phonetic phenomenon of → metrics (→ Metron; → Metrical Laws) is a kind of spontaneous sound change which naturally originates in oral language and affects a word-final and a word-initial vowel, the latter immediately following the former, in causing a contraction of both, but with different results (→ Vowel Changes). This sound change is restricted to words which are syntactically connected. It is not to be confused with other phonetic phenomena like → elision, → aphaeresis and synaloephe, respectively, all of which happen under the same circumstances, i.e. when a word-initial vowel immediately follows a word-final one, and are clearly denoted in writing; any → contraction which happens over the 'gap' between two words and does not become visible in writing, is called synizesis, just like any other invisible contraction which happens within a single word (Schwyzer 1939:401). With crasis both vowels are contracted into one long vowel, the color of which, according to the contraction rules (de Haas 1988, Fritz 2003:812f.), is normally different from the timbres of the original vowels respectively, e.g. *ho ex* > *hoûx* 'the one from', *ho áristos* > *hṓristos* 'the best', *tò ónoma* > *toúnoma* 'the name'; if the second word starts with a → diphthong containing an [i], the resulting vowel is long, with an iota subscript also spelled in writing, *egṑ oîmai* > *eg[ǫ́]mai* 'I opine' (ἐγῷμαι). If the word-final vowel is preceded by a consonant, there is a specific diacritical sign to mark the vowel which is the result of a contraction caused by crasis: that is called 'coronis' (*korōnís* 'curved'). This specific diacritic used to have its own form, but in modern editions, it looks like the sign for smooth breathing (*spiritus lenis*): *tà álla* > *t[ắ]lla* 'the other-neut. pl.'. Crasis may also take place if the second vowel does not follow the first vowel immediately, but there is a word-final → semivowel /i/ in between: that semivowel belonged originally to the first syllable as the non-syllabic constituent of a diphthong, but it was eventually eliminated due to its intervocalic position, i.e. it was jettisoned before the contraction of the two vowels took place, which only after the deletion of the /i/ came into direct contact with each other: **ei án* > *ến* 'if', *kaì egṓ* > *kágṓ* 'and I'. Crasis may even take place when the second vowel is preceded by rough breathing (*spiritus asper*), which is maintained after contraction, and depending on the type of the consonant which precedes the first vowel, it may lead to the → aspiration of that consonant: *tà hópla* > *thṓpla* 'the arms', *tò himátion* > *thoímátion* 'the cloak'. These examples, which demonstrate a combination of an aspirated consonant with what superficially looks like the sign for smooth breathing (*spiritus lenis*), may be taken as a proof that this sign is actually not what it looks in the first place. In some cases where crasis is expected, the second word, if it is more meaningful than the first one, preserves its word-initial vowel unchanged, while the word-final vowel of the preceding word disappears; that phenomenon of external vowel deletion would be the equivalent of hyphaeresis, which is restricted to the interior of a word: e.g. *ho anḗr* > *hánḗr* 'the man', *toû andrós* > *tándrós* 'of the man', *tò autó* > *taùtó* 'the same-neut.

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Osthoff's Law

"Osthoff's Law" is a phonological process within the history of Greek (and some other IE languages) by which long → vowels ("V̄") were shortened when they occurred before sonorants ("R", i.e. liquids, nasals, or glides) that were followed by another → consonant ("C"): -V̄RC(-) > -V̆RC(-) (Lejeune 1972:219–220, Rix 1992:56, Sihler 1995:58–59, 74). The sonorant, moreover, must occur in the coda position of → syllables, and not in the onset (Kobayashi 2004:26, Miller 2010:187). Thus the thematic dat. pl. ending Gk. *-ois* < PIE **-ōis* (cf. Skt. *-ais*, with long diphthong) shows the result of this shortening (*/-ōys/ > /-oys/); but the long vowel of Gk. *krēmnós* 'cliff, overhang' (assuming syllable division /krē.mnos/, cf. word-initial *mn-*: Gk. *mnêma* 'tomb' etc.) is not affected. The original analysis of material in Greek and elsewhere was proposed by the Indo-Europeanist Hermann Osthoff (1847–1909) in 1879; for details, with a survey of initial reactions and ensuing theoretical discussions through the 1970's, see Collinge 1985:127–131.

The similar shortening processes elsewhere in IE (as in Italic, Celtic, Germanic, Balto-Slavic), also often referred to as "Osthoff's Law" (or "Osthoff-shortenings"), are independent of the Greek version of Osthoff's Law (Fortson 2010:70–71): such a shortening process is a typologically natural strategy for the avoidance of "overlength" (or of "superheavy" syllables (→ syllable weight)), a phonological tendency that is manifested in many languages in a variety of ways (Devine & Stephens 1994:78, Gordon 2006:51; on the typology of the Osthoff's Law context specifically, Gordon 2006:234). The same tendency is also an important feature of Greek, Sanskrit, and Proto-Indo-European metrical composition (Hoenigswald 1989, 1991).

Besides the thematic dat. pl. (above), Osthoff's Law appears in a number of isolated words, including early borrowings (e.g. Iran. **Pārsa-* 'Persian' > pre-Ion. **Pērsa-* > Att., Ion., Dor. *Persai* 'Persians'), in the nom. sg. of nouns in *-eus* (< **-ēus*, with stem **-ēw-*, cf. Hom. gen. sg. *-ē(w)os*), and in participles based on long-vowel stems (e.g., aor. passives in *-(th)ē-* (→ Aorist Formation) make participles in *-(th)e-nt-*, not ***-(th)ē-nt-*). But in many cases, → analogy has restored or maintained the long-vowel form: e.g., 3 pl. pres. subj. mid. *-ōntai* (cf. 1 sg./pl. *-ōmai*/*-ṓmetha* → Present Tense); post-contraction nom. pl. pres. ptc. *tīmôntes* 'honoring' (< **tīmáontes*); augmented forms of verbs with initial diphthong (e.g. imperf. *ōídeon* to *oidéō* 'swell'); Hom. (Ion.) dat. pl. *nēusí* (cf. other case forms in *nē-*: acc. sg./pl. *nêa/nêas*, gen. sg./pl. *nēós/nēôn*, etc.).

Given the phonetic naturalness of Osthoff-shortening, the process occurred not only pandialectally at an early period, but also later. Thus some words with onset **eweRC-* underwent *w*-loss and vowel → contraction, resulting in **ēRC-*, followed by a late Osthoff-shortening in Ionic: cf. **ewelde/o-* 'long for' (> Hom. *eéldomai*) > **ēlde/o-*, whence (with Osthoff-shortening) Hom. *éldomai* (Nussbaum 1998:143). (Similarly Peters 1980:228–286 on Gk. *-kraira* '-horned, -headed'.)

The absolute chronology of the pandialectal Osthoff-shortening is controversial, in part because of complexities in evaluating the relative chronology of Osthoff's Law and other phonological processes (especially medial cluster simplifications involving resonants), and also because of indeterminacies in → Mycenaean Greek orthography, where (for example) vowel length is not notated. For discussion, see Lejeune 1972:220, Peters 1980:306–319.

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Leslie Threatte

Schwa Secundum

'Schwa secundum' refers to an epenthetic vowel inserted to break up certain word-initial consonant clusters in Proto-Indo-European (Fortson 2010:68). In the standard version of this theory (Schindler 1977:31, Mayrhofer 1986:175–177), the word-initial sequence #(s)TTRV- is adjusted to #(s)T$_e$.TRV-, where T = obstruent, R = resonant (i.e., liquid, nasal, or glide), V = vowel (or other syllabic segment), period indicates → syllable boundary, and subscript "e" indicates the position of the inserted vowel (although other notations are also used, such as subscript "ə" [schwa] or superscript "o"). Such sequences tend to occur in PIE zero grades (→ Ablaut) (i.e., root forms with no vowel at all), where #(s)TT- is the zero grade of a PIE root, and RV- corresponds to suffixal material. The best-known example involves the PIE word for the number '4': its initial syllable had *e*-grade vocalism in some forms, as in nom. pl. **k^{w}etwores* (> Att. Gk. *téttares*, Old Church Slavic *četyre*, Welsh *pedwar*, etc.); but other forms had zero grade in the first two syllables, as in loc. pl. **k^{w}tw̥rsu*, which thus displays the typical schwa secundum context. The regular result of schwa secundum in Greek is /i/ (Lejeune 1972:208), hence the *i*-vocalism of Hom. (Aeol.) *písures* '4' (nom. pl., but based on the zero grade **k^w$_e$.tw̥r-*). Another example is the nasal-infixed (→ Nasal Presents) verb stem PIE **pt-ne-h$_2$-* 'spread' > **p$_e$.tneh$_2$-*, whence Gk. *pítnēmi* 'spread out'. (The regular result of schwa secundum in Italic is /a/, hence the Latin cognates: Lat. *quattuor* '4', *pandō* 'spread out'.) Examples like *pítnēmi* (*Iliad* and later) and (with similar structure and background) *skídnamai* 'be scattered' (*Iliad* and later) led to the analogical extension of *i*-vocalism in the initial syllables of forms with similar morphology, e.g. *kírnēmi* 'mix' (*Odyssey* and later), *krímnēmi* 'hang' (Pindar and later).

The original version of the theory (Güntert 1916), formulated in terms of a poorly-defined PIE 'reduced grade', was highly unconstrained; the theory was designed to address many problems of Indo-European vocalism that have now been solved in other ways, and most of Güntert's proposals have been abandoned. (On the term 'schwa secundum', as opposed to 'schwa primum', see Mayrhofer 1986:176-177.)

Forms like *hidrúō* 'establish' (< **s$_e$.dru-ye/o-*, based on the zero grade of PIE **sed-* 'sit') show that for the schwa secundum context in Greek, the initial "T" (in the above formula) is not restricted to stops. It has also been suggested (Vine 1999) that the same process accounts for the *i*-vocalism of some Gk. forms with other environments, e.g., *rhíza* 'root' (Myc. *wi-ri-za*) < (oblique stem-form) **wr$_e$.dyV-* and *rhímpha* 'swiftly' < **wr$_e$.nkhwa.* There may also be instances in which the process applied, within Greek, to medial sequences -TRRV- (> -T$_e$.RRV), as in the fem. → agent noun allomorph Myc. *-ti-ra$_2$* /-tirra/, Thess. *-tarra* (< **-terra*), Hom. (Aeol.) *-teira* (all ultimately from **-t$_e$.rya* < PIE **-tr-ih$_2$*) and Att. *oiktírō* 'have pity' < Proto-Gk. **oikt$_e$.rye/o-*, denominative to *oiktrós* 'pitiable'. (See on these forms Hajnal 2007:151–152.) The morphological background of Gk. *rhíza* and the inner-Greek word-medial examples suggest that the PIE vowel-insertion processes persisted into the development of the individual IE languages.

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